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Tactical Stealth
Visby as a Case Study

Naval Postgraduate School

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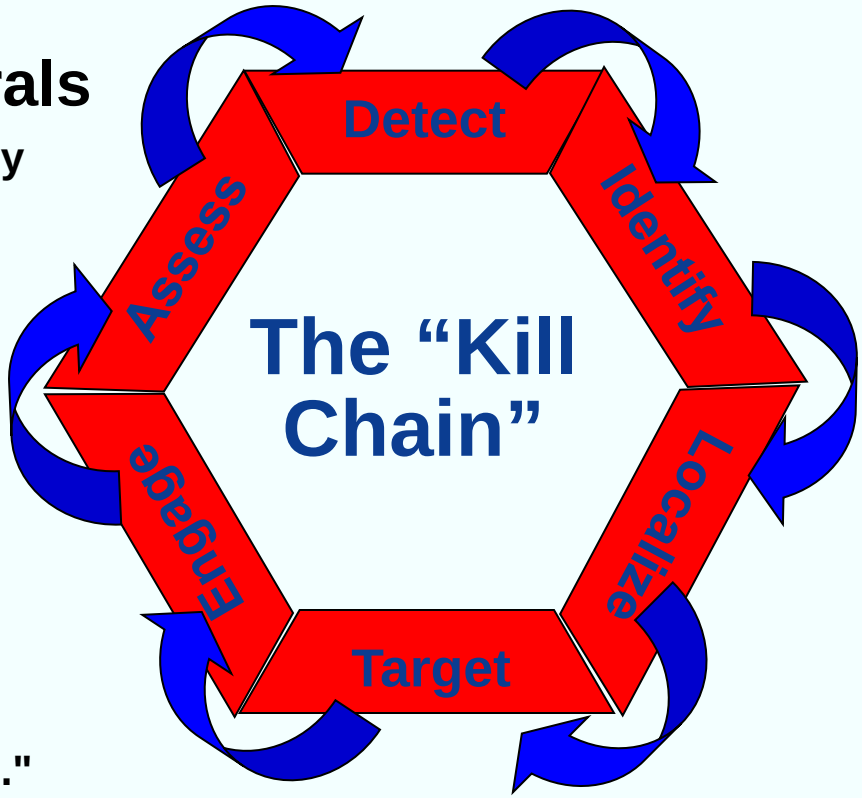
What are the Risks? Where Are the Risks?

- **In Combat, How Are Platforms at Risk?**
  - What are the hallmarks of “harm’s way”?
- **Risk is Most Pressing in Littorals**
  - " will be required to operate in the relatively shallow water of littoral regions where the threat posed by mines, patrol combatants, small boats, submarines armed with wake homing torpedoes, sea- and land-based aircraft (fixed and rotary wing), coastal defense sites (artillery, missile and possibly torpedo), and terrorist groups can be significant."
- **Comments**
  - "Integration of full spectrum signature reduction, active and passive self defense systems, and physical ship design is critical."



The Swedish Experience

- **Defense Posture**
  - Historical / Cultural Focus
    - 2,000 Miles of Coastline (3,218 Kms)
    - Historical Conflicts with Russia, France, Denmark, Prussia, Finland
    - Plus Contemporary Demands of Participation in European Politics
    - Coalition Forces, Joint Operations
- **Defense Objectives**
  - Defend Sweden against armed attack
  - Maintain Swedish territorial integrity
  - Contribute to peace and security in the world
  - Assist Sweden in times of severe peacetime difficulty
- **Requires Swedish FMV to Procure Dual-Use Weapon Systems**
  - "Our aim should be to create capabilities that are suited for international tasks ***and*** national territorial defense tasks"

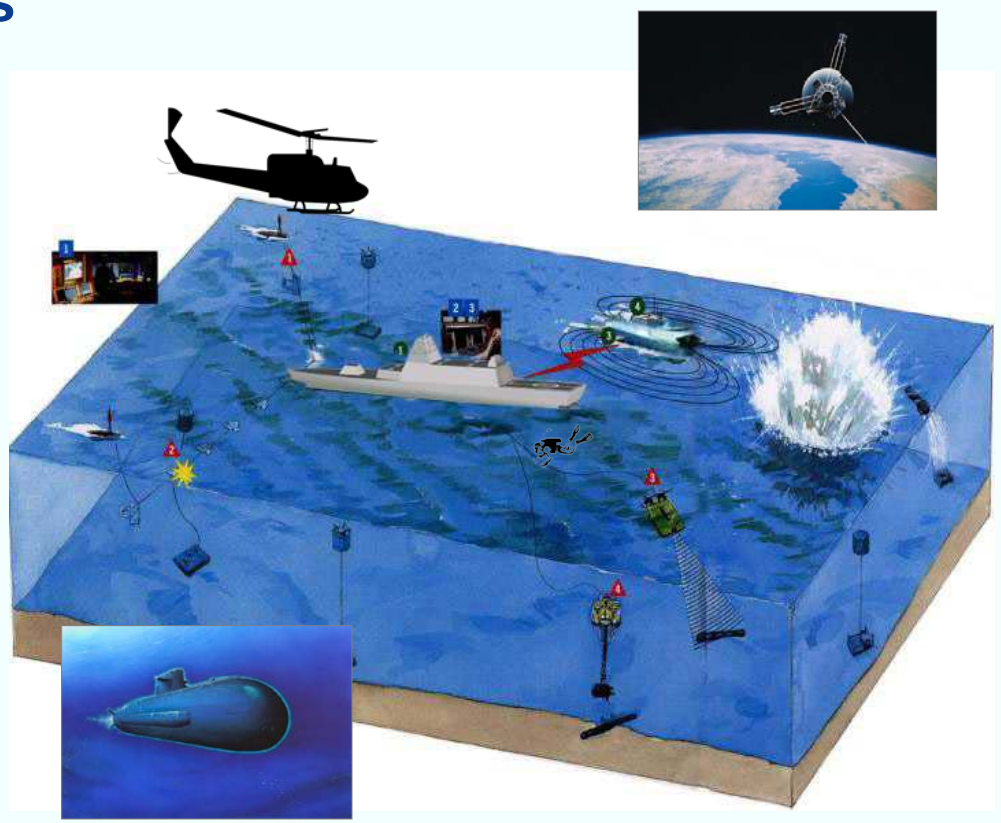
    GEN Håkan Syrén, Supreme Commander of Swedish Armed Forces, January 21, 2004

Equates to a *Littoral* Focus

The Littoral Challenge

The Spectrum of Signatures

- Radar cross-section (RCS)
- Infrared signature (IR)
- Acoustic signature (hydroacoustic and airborne noise)
- Magnetic signature
- Underwater electrical potential (UEP)
- Pressure signature
- Visual signature
- Transmitted signals
- Laser cross-section



Answering the Challenge

- **Management of Radar Cross Section**
  - **Shaping, Composites, Concealed installation, Topside Integration, Embedded Antennas**
    - **Frequency Selective Surfaces (FSS)**
    - **Limited use of Radar Absorbent material (RAM)**
    - **RCS modeling**

Vosper Thornycroft: Sea Wraith

DCN International: CCX 21

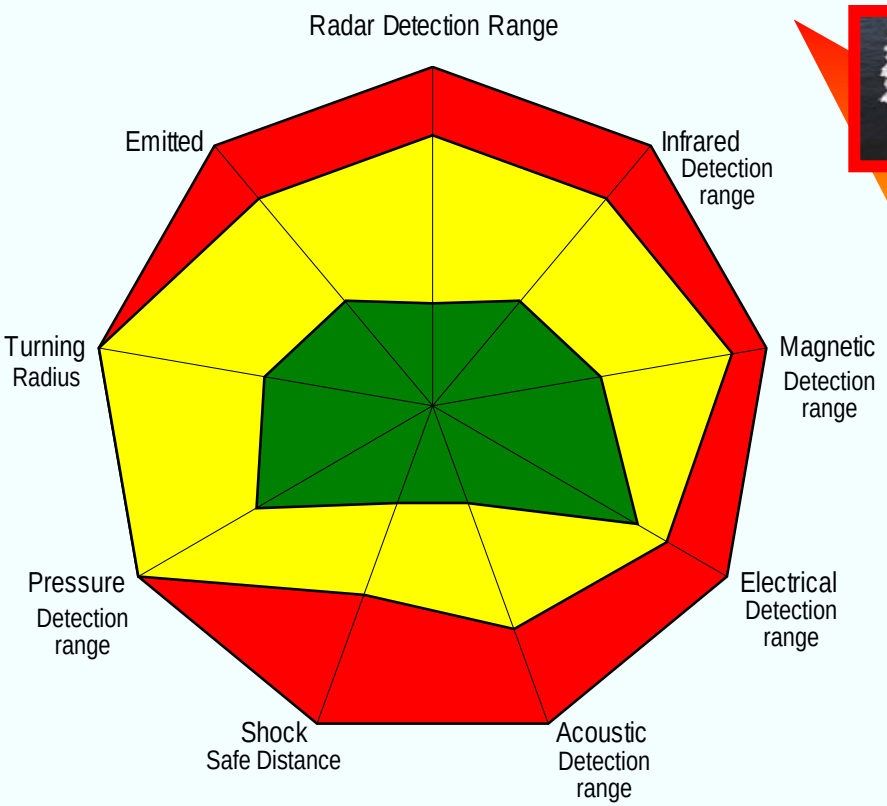
DCN International: La Fayette

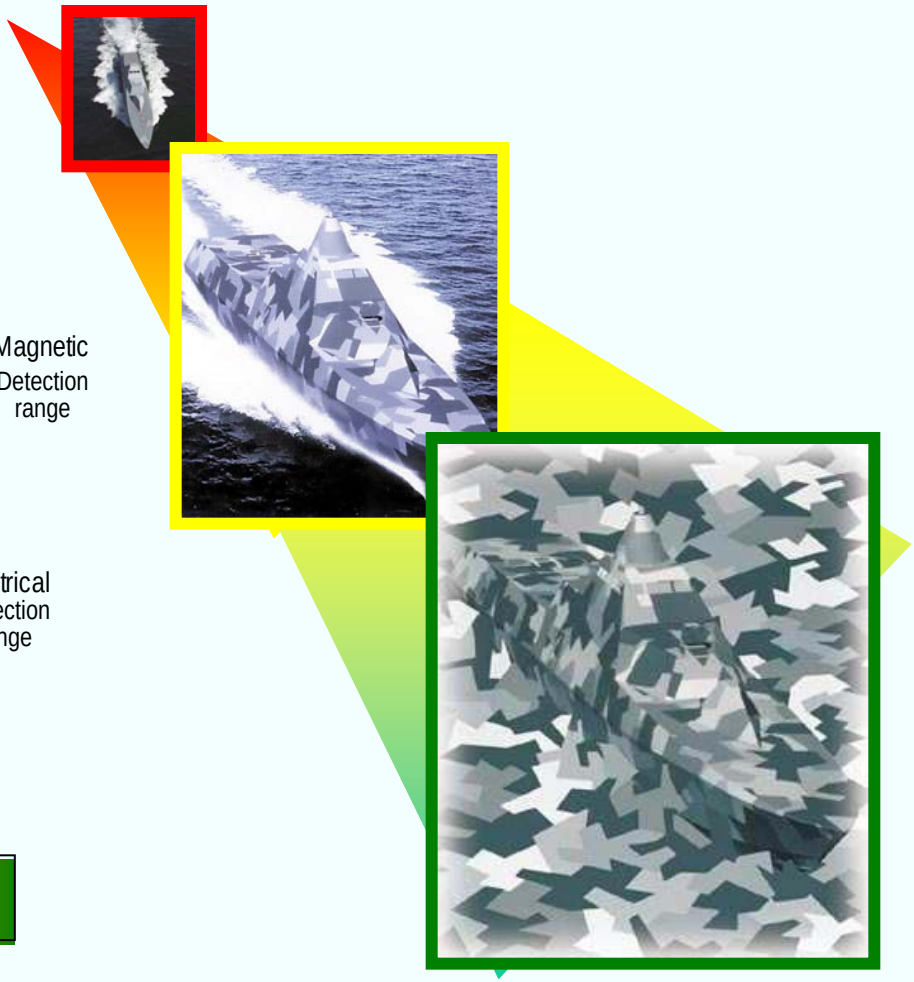
Kockums: Smyge

Greatest Progress to Date
in a Small Combatant

- **Swedish Visby Class**
  - **Length 73 m / Beam 10.4 m / Displacement 600 m tons**
  - **Sandwich construction carbon fiber reinforced plastic (CFRP)**
  - **4 x gas turbines + 2 x diesel engines / 2 x waterjet propulsors / >35 knots**

Relative Tactical Superiority of RSN Visby





Representative Tactical Analysis

Other Surface Ship

Operating Area

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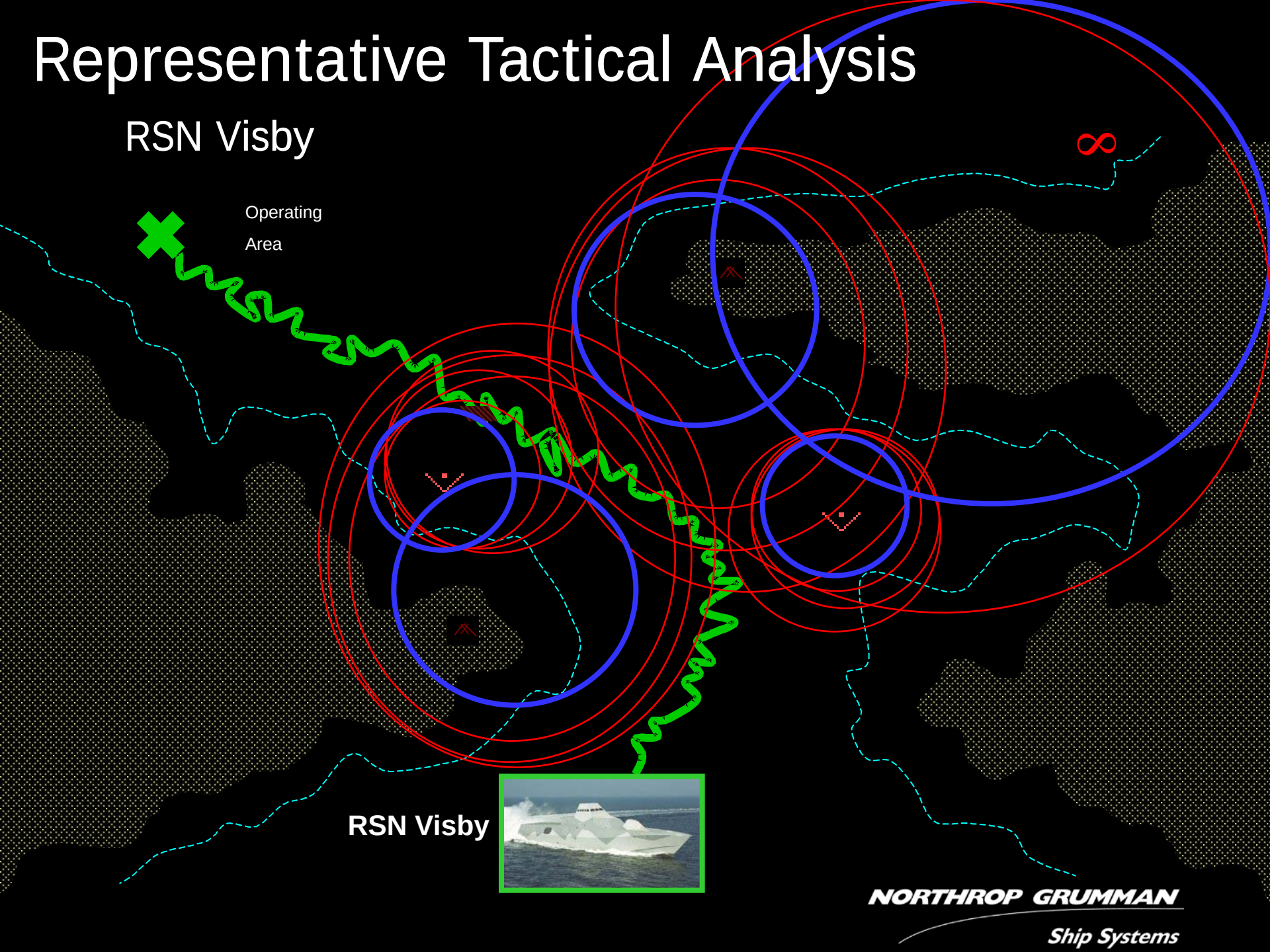
Other Surface Ship
HMS Triton, a trimaran

NORTHROP GRUMMAN
Ship Systems

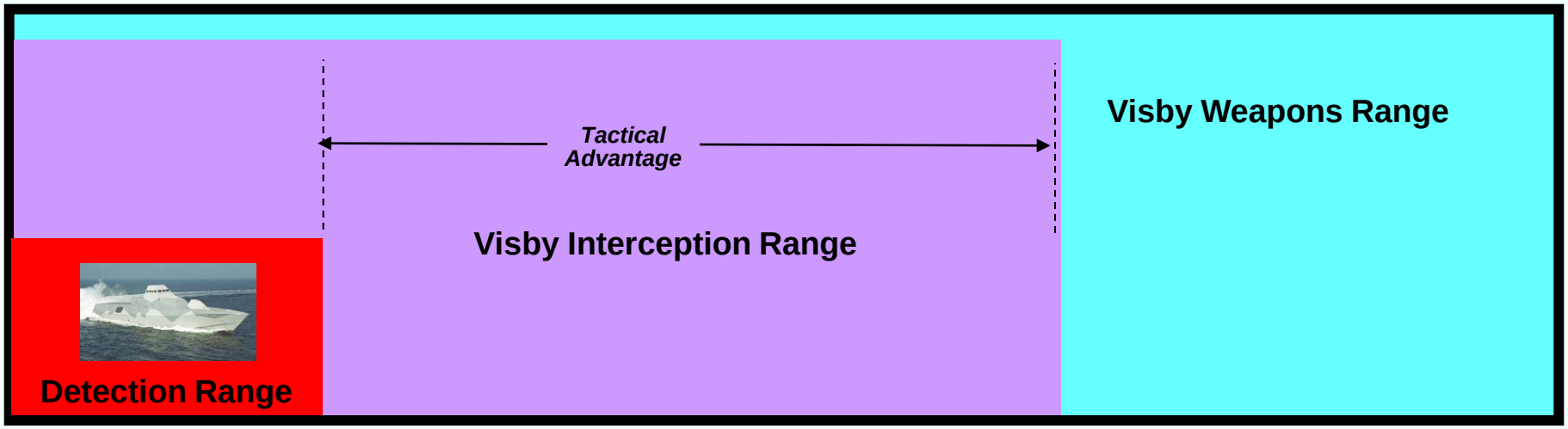
Representative Tactical Analysis

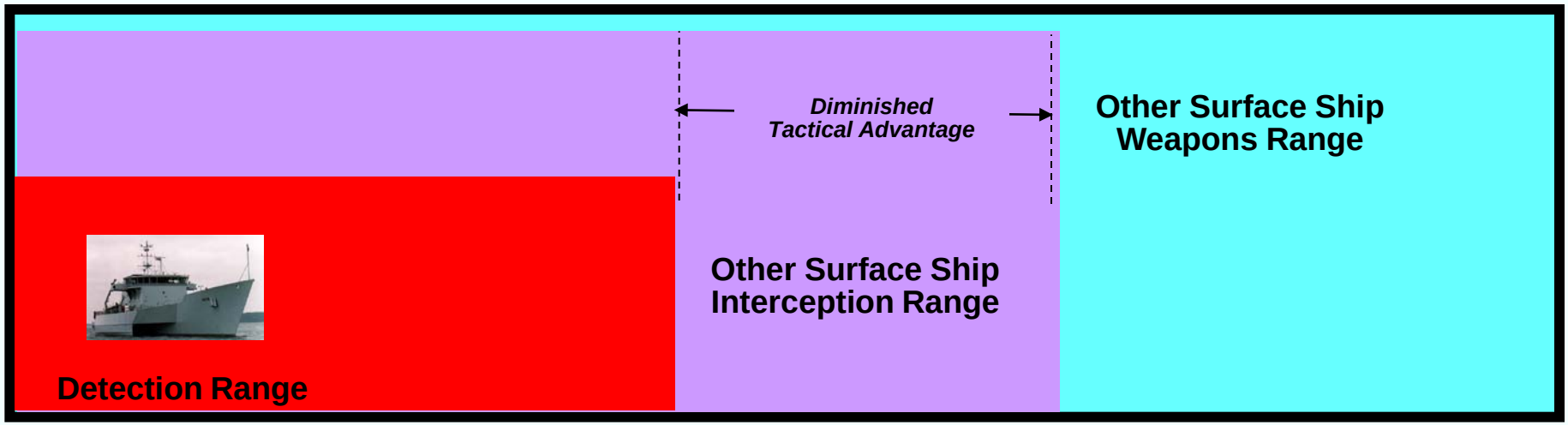
RSN Visby



NORTHROP GRUMMAN
Ship Systems

Representative Tactical Analysis





Challenge to Systems Engineers

- Responding to Operational Requirements
- Conducting Conceptual Analysis of Alternatives
- Allocation of Top Level Requirements
- Conducting Specific Trade Off Studies
- Balancing Cost, Schedule & Performance

Tactical Stealth Through RCS Management
NGSS Contributions

- **USS Radford, DD 968**
  - Fitted with Advanced Enclosed Mast / Sensors (AEM/S), July 1997
  - ONR Funded Technology Demonstration Project
- **USS San Antonio, LPD 17**
  - Fitted with 2 x AEM/S
  - Rapid Transition from R&D to New Construction
  - USS San Antonio to be delivered to the fleet in November

USS Radford, DD 968

USS San Antonio, LPD 17

- **The Littorals are *harm's way***
- **Full Spectrum Signature Management Essential to War Fighter's Success**
- **Systems Engineering is *the* Enabling Discipline**

- **Swedish Visby is Small Combatant Benchmark for Our Own US Navy**